Transferring Complex-Domain Representations to Real-Valued Networks through Knowledge Distillation

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Abstract

This study proposes a Complex-valued Knowledge Distillation framework. The framework is designed to improve the efficiency of lightweight neural networks in processing complex-valued wireless signals. The proposed method transfers the learned representations from a complex-valued teacher network (CVNN) to a real-valued student network (CNN). Experimental results show that the distilled CNN consistently outperforms both the baseline CNN and the teacher CVNN across all SNR conditions. In particular, the proposed method improves performance by up to 12.23% in low-SNR environments. This confirms the robustness and efficiency in preserving complex-domain characteristics under real wireless channel conditions.

1 Introduction

Deep learning (DL) has gained attention in signal processing for its ability to extract features directly from raw signals. Convolutional Neural Networks (CNN) are widely used because CNN effectively learn structural patterns in signals [1]. However, most CNN are designed for real-valued inputs, while real-world signals are complex-valued. This mismatch causes incomplete representation of correlations between real and imaginary parts, leading to information loss and reduced performance. To address this limitation, Complex-Valued Neural Networks (CVNN) have been proposed [2]. CVNN jointly learn real and imaginary components, preserving amplitude–phase relationships and improving complex signal representation. However, a large number of parameters and high computational cost limit the practical deployment of CVNN. To overcome this issue, this study applies Knowledge Distillation (KD) to transfer knowledge from a high-capacity CVNN teacher model to a lightweight CNN student model. The CNN student learns from the teacher, achieving stable and efficient performance

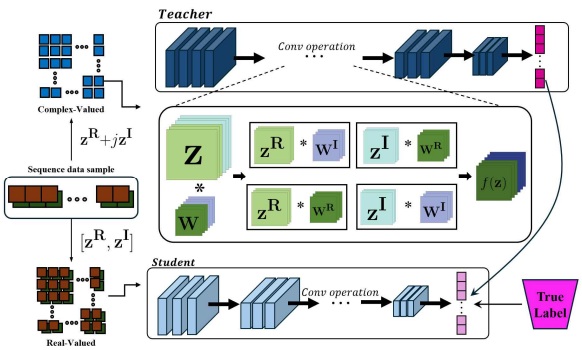

Figure 1 Complex-valued Knowledge Distillation framework

2 Complex-valued Knowledge Distillation Framework

CVNN is a neural network that directly processes complex-valued inputs composed of real and imaginary parts. It was proposed to overcome the limitations of conventional real-valued CNN. The input data X and filter W are represented as follows, consisting of their real and imaginary components:

$$f(\mathbf{z}) = \mathbf{W}^{\mathbf{R}}\mathbf{z}^{\mathbf{R}} - \mathbf{W}^{\mathbf{I}}\mathbf{z}^{\mathbf{I}} + \mathbf{j}(\mathbf{W}^{\mathbf{I}}\mathbf{z}^{\mathbf{R}} + \mathbf{W}^{\mathbf{R}}\mathbf{z}^{\mathbf{I}}) \qquad (1)$$

where z denotes the weight matrix and the complex-valued signal. R and I indicate the real and imaginary parts. f denotes a complex-valued operation. CVNN

preserves interactions between the real and imaginary components.

The Complex Knowledge Distillation method aims to transfer knowledge from a complex-valued teacher network to a real-valued student network. Through the teacher's output responses, the student acquires key complex-domain characteristics, including phase and amplitude information. This learning process enhances the student's representational capacity while ensuring compactness and computational efficiency. Loss function reduces the distributional gap between teacher and student outputs, allowing the student to maintain complex-domain expressiveness with a lightweight and efficient structure.

$$L_{KD} = \tau^2 \cdot KL(p_s || p_t) \quad (2)$$

Here, p_s and p_t denote the softmax outputs of the teacher and student models, respectively. The temperature parameter τ controls the smoothness of the probability distribution. Here, $KL()$ denotes the Kullback–Leibler divergence, which measures how the student model probability distribution diverges from the teacher model. Through this process, the knowledge learned by the complex-valued teacher model can be efficiently transferred to the real-valued student model. This reduces computational complexity while preserving the characteristics of complex signals.

3 Experiments

In this study, a 5G software modem–based testbed was built to collect real-world 5G wireless signals [3]. It was used to evaluate performance under different SNR conditions. Each complex-valued sample consists of 144 real and 144 imaginary components labeled by eight modulation indices. Knowledge distillation was performed between a CVNN teacher and a CNN student with the same network architecture. This experiment evaluated the transferability of complex-domain representations to a lightweight real-valued model.

	Accuracy (%)		
SNR (dB)	**CNN**	**CVNN**	**Distilled CNN**
-2.51	96.70	96.90	96.70
-2.74	96.60	96.45	**97.03 (+0.43 ↑)**
-2.81	96.28	95.79	**97.24 (+0.96 ↑)**
-2.99	93.93	94.66	**95.60 (+1.67 ↑)**
-3.13	92.08	92.78	**93.88 (+1.95 ↑)**
-3.42	88.41	89.68	**90.28 (+1.87 ↑)**
-3.70	76.08	77.45	**80.58 (+4.50 ↑)**
-4.11	42.35	49.73	**54.58 (+12.23 ↑)**

Table 1 Accuracy comparison across SNR levels

The experimental results demonstrate that the CNN model trained through knowledge distillation consistently outperformed the CVNN across all SNR conditions. This indicates that the proposed distillation framework effectively transfers the complex-domain knowledge of the teacher model to the real-valued student model. Notably, in low-SNR environments, where signal degradation is more severe, the distilled CNN achieved up to a 12.23% improvement in accuracy compared to the baseline CNN. These results confirm that the proposed method enhances the robustness and generalization capability of the lightweight student model while maintaining high performance under challenging channel conditions.

4 Conclusion

This study proposes a Complex Knowledge Distillation method that transfers knowledge from a complex-valued teacher model to a real-valued student model while preserving amplitude–phase characteristics. Experiments demonstrate higher robustness and accuracy than conventional models, and future work will extend the approach to diverse complex-valued datasets and signal processing tasks to further validate its generalization.

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